

## MatSci 465: Advanced Electron Microscopy & Diffraction

### Project Proposal

*Due: Friday, February 13, 2026*

**Project Title:** Analyzing the effects of strain on a Supervised Machine Learning Model for Pseudosymmetry correction in EBSD

**Team Members:**

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**Date:** 02/15/26

1. **Project Objectives:** What problem will you address? What are your specific goals?
2. **Methods & Approach:** What computational techniques, algorithms, or frameworks will you use
3. **Dataset Description:** What data will you use? (Course datasets, public repositories, or your own data?) Include data type, size, and source.
4. **Expected Outcomes:** What results do you anticipate? How will you evaluate success?

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### Proposal (250 words)

Electron Backscatter diffraction (EBSD) is an important characterization technique to analyze a material's microstructure and crystalline orientations. EBSD utilizes Kikuchi patterns to index the collected diffraction pattern. Most orientation determination software has high precision, however complex materials can experience pseudosymmetry problems, caused by similar/nearly identical crystallographic orientations. This can lead to incorrect determination of the crystal orientation. In addition, added strain in a material can enhance the pseudosymmetry in a material. Our project plans to analyze and understand how the accuracy of determining crystallographic orientations changes as strain increases. A supervised learning model will be developed to predict the amount of strain from an inputted EBSD image for a targeted material. Utilizing the Materials Project and VESTA an unstrained crystallographic information file (cif) that contains the information needed will be made. EBSD patterns will be simulated using EMsoft software. Strained EBSD patterns will be generated with a known amount of strain. Once generated, strained EBSD images will be randomly split into training and testing bins. The training data will be labeled and fed into our model over multiple rounds. Between rounds, the model will be tested with the testing data, with a known amount of strain but unlabeled. The model will be evaluated using precision, recall, and f1 score metrics. The percent error from the known value will be tabulated, and the standard deviation for each round of testing will be recorded and compared. The model will then be evaluated for over fitting and under fitting and iteratively adjusted.

## Dataset Information

### Dataset Source:

<https://github.com/pyxem/kikuchipy-data>

[https://kikuchipy.org/en/stable/user/open\\_datasets.html](https://kikuchipy.org/en/stable/user/open_datasets.html)

Example: <https://zenodo.org/records/5803073>

**Data Type:** Electron Backscatter Diffraction (EBSD) diffraction patterns

**Dataset Size & Format:** .bmp, cif, ctf

**Preprocessing Requirements:** Simulation EBSD patterns from cif files and possible filtering of EBSD patterns

**Ethical/Proprietary Considerations:** Materials Project, VESTA, and EMsoft are open source.

## Preliminary References

- [1] B. E. Jackson, S. I. Wright, and A. J. Wilkinson, "Resolving pseudosymmetry in  $\gamma$ -TiAl using cross-correlation electron backscatter diffraction," *Ultramicroscopy*, vol. 194, pp. 84-95, 2018.
- [2] C. Maurice, K. Dzieciol, and R. Fortunier, "A method for accurate localisation of EBSD pattern centres," *Ultramicroscopy*, vol. 111, no. 2, pp. 140-148, 2011.
- [3] Z. Ding and M. De Graef, "Parametric simulation of electron backscatter diffraction patterns through generative models," *npj Computational Materials*, vol. 9, art. 211, 2023. [Online]. Available: <https://doi.org/10.1038/s41524-023-01143-z>
- [4] M. De Graef *et al.*, "EMsoft: Open-source software for electron diffraction/microscopy simulations," GitHub Repository, 2024. [Online]. Available: <https://github.com/EMsoft-org/EMsoft>
- [5] H. W. Ånes *et al.*, "Pattern matching tutorial," *kikuchipy Documentation*, 2024. [Online]. Available: [https://kikuchipy.org/en/stable/tutorials/pattern\\_matching.html](https://kikuchipy.org/en/stable/tutorials/pattern_matching.html)
- [6] K. Kaufmann, D. Maryanovsky, W. M. Mellor *et al.*, "Efficient few-shot machine learning for classification of EBSD patterns," *npj Computational Materials*, vol. 4, art. 106, 2023. [Online]. Available: <https://doi.org/10.1038/s41524-023-01143-z>
- [7] K. Kaufmann, C. Zhu, A. S. Rosengarten, and D. Maryanovsky, "Crystal symmetry determination in electron diffraction using machine learning," *Science*, vol. 367, no. 6477, pp. 564-568, 2020.
- [8] M. K. Horton, P. Huck, R. X. Yang *et al.*, "Accelerated data-driven materials science with the Materials Project," *Nature Materials*, vol. 24, no. 10, pp. 1522–1532, Jul. 2025.
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